Software Requirements Specification (SRS)

Project Name: KidDose / DoseDek (Pediatric Dosage & Safety Calculator)

Document Version: 1.0.0

Status: Draft / System Specification

Standard: ISO/IEC/IEEE 29148 / IEEE 830 Compliant

1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) document describes the complete functional, non-functional, and interface requirements for the **KidDose / DoseDek** mobile application. It serves as the definitive reference for developers, quality assurance engineers, medical content reviewers, and project stakeholders.

1.2 Scope

KidDose / DoseDek is a cross-platform (iOS and Android) offline-first mobile application designed to assist healthcare professionals (physicians, interns, pediatricians, pharmacists, emergency staff) in calculating pediatric medication dosages accurately and instantly.

Key capabilities include:

- Real-time dosage calculation based on a single patient weight input (sub-100 ms rendering time).
- Automated safety checks including age contraindications, disease-specific alerts (e.g., G6PD deficiency, renal impairment), and daily/single dose caps.
- Offline-first local database execution with bundled drug data.

1.3 Definitions, Acronyms, and Abbreviations

- **SRS:** Software Requirements Specification
- **TDD:** Technical Design Document
- **PRD:** Product Requirements Document
- **G6PD:** Glucose-6-Phosphate Dehydrogenase Deficiency
- **OPD:** Outpatient Department
- **ER:** Emergency Room
- **Max Cap:** The maximum allowable dose (single or daily) enforced to prevent toxicity.
- **UI/UX:** User Interface / User Experience

1.4 Document Overview

The remainder of this document is organized into:

- **Section 2:** Overall Description (product perspective, user classes, operating environment).
- **Section 3:** Functional Requirements (detailed behavior by feature module).
- **Section 4:** External Interface Requirements (UI, hardware, software).
- **Section 5:** Non-Functional Requirements (performance, safety, security, quality attributes).
- **Section 6:** Verification & Acceptance Criteria.

2. Overall Description

2.1 Product Perspective

KidDose is a standalone, client-side mobile application with zero runtime server dependency for core calculations. It replaces manual reference books, paper dosage tables, and multi-step digital calculators with a single-input, highly responsive interface.

```
+------------------------------------------------------------+
|                     CLIENT MOBILE APP                      |
|                                                            |
|  +-------------------+  +----------------------------------+  |
|  |  User Input (kg)   |  |  Local Bundled DB (Isar/SQLite)    |  |
|  +-------------------+  +----------------------------------+  |
|            |                            |                  |
|            v                            v                  |
|  +-----------------------------------------------------+  |
|  |             In-Memory Calculation & Safety Engine       |  |
|  +-----------------------------------------------------+  |
|                              |                             |
|                              v                             |
|  +-----------------------------------------------------+  |
|  |     Instant UI Rendering (mg, mL, Red Flag Highlights)    |  |
|  +-----------------------------------------------------+  |
+------------------------------------------------------------+
```

2.2 Product Functions Overview

1. **Weight & Patient Context Management:** Capture weight, age, and clinical risk flags.
2. **Dynamic Dose & Volume Calculation Engine:** Calculate target mg and liquid volume (mL).
3. **Automated Safety Evaluation:** Validate age limits, contraindications, and dose caps.
4. **Preset Category Navigation:** Filter drugs by clinical context (OPD, ER, Asthma, Antibiotics).
5. **Concentration Switching:** Toggle available liquid preparations without navigating away.
6. **Local Persistence:** Retain user preferences and favorite pinned drugs locally.

2.3 User Classes and Characteristics

- **General Practitioners / Interns:** High workload, fast decision-making environment. Require maximum speed, zero friction, and clear safety guardrails.

- **Pediatric Specialists:** Require high precision, access to second-line drugs, and advanced dosage adjustments.
- **ER Doctors / Nurses:** Require rapid access to emergency/resuscitation drug protocols during critical resuscitation events.
- **Pharmacists:** Require verification of concentration matching and maximum daily dose safety limits.

2.4 Operating Environment

- **iOS:** iOS 14.0 or higher.
- **Android:** Android 8.0 (API Level 26) or higher.
- **Connectivity:** 100% operational without network connection. Optional Wi-Fi/Cellular connectivity for database updates only.

2.5 Design & Implementation Constraints

- **Platform:** Developed using Flutter (Dart framework).
- **Footprint:** App binary package size must not exceed 30 MB.
- **Data Privacy:** Zero storing or transmitting of Patient Health Information (PHI/PII). All calculation inputs are transient in memory.

3. Specific Functional Requirements

3.1 Module 1: Patient Context & Weight Input

- **FR-1.1:** The app **SHALL** provide a prominent single numeric input field for Patient Weight in kilograms (kg) on the primary dashboard.
- **FR-1.2:** The weight input **SHALL** accept floating-point numbers with up to 2 decimal places (e.g., 8.45 kg).
- **FR-1.3:** The weight input field **SHALL** automatically gain focus or accept rapid numpad entries without requiring modal dialogs.
- **FR-1.4:** The app **SHALL** provide optional secondary input controls for Patient Age (in months or years).
- **FR-1.5:** The app **SHALL** provide quick toggle switches for clinical risk factors:

    - `[G6PD Deficiency]` (Boolean: On/Off)
    - `[Renal Impairment]` (Boolean: On/Off)
- **FR-1.6:** Changing any input field **SHALL** immediately update all rendered drug results in $\le 100\text{ ms}$.

3.2 Module 2: Dosage & Volume Calculation Engine

- **FR-2.1:** For a given drug and patient weight W (kg), the engine **SHALL** calculate the raw single dose D_{raw} (mg) as:
  $$D_{raw} = W \times \text{Dose}_{std}$$

  where Dose_{std} is the standard dosage parameter (mg/kg/dose).
- **FR-2.2:** The engine **SHALL** clamp the calculated dose D_{final} (mg) against the drug's Maximum Single Dose Cap M_{single} (mg):
  $$D_{final} = \min\left(D_{raw}, M_{single}\right)$$
- **FR-2.3:** The engine **SHALL** calculate the liquid volume V (mL) based on selected concentration parameters C_{mg} (mg) per C_{ml} (mL):
  $$V = \frac{D_{final}}{C_{mg}} \times C_{ml}$$
- **FR-2.4:** Liquid volume results **SHALL** be rounded to 1 decimal place (e.g., $3.75\text{ mL} \rightarrow 3.8\text{ mL}$) for clinical clarity.
- **FR-2.5:** The engine **SHALL** evaluate total daily dosage D_{daily} against Maximum Daily Dose Cap M_{daily} (mg):
  $$D_{daily} = D_{final} \times N_{doses}$$

  If $D_{daily} > M_{daily}$, the app **SHALL** flag a "Daily Dose Exceeded" warning.

3.3 Module 3: Real-Time Safety & Red Flag Rules

- **FR-3.1 Age Limit Check:** If $Age_{patient} < Age_{min\_drug}$, the drug card **SHALL** display a prominent **RED ALERT** banner stating: *"Contraindicated in patients under X months/years"*.

- **FR-3.2 G6PD Check:** If `[G6PD Deficiency]` is toggled **ON** and $Drug.g6pdSafe == \text{false}$, the app **SHALL** visually flag the drug with a **RED WARNING** tag and disable standard volume display until acknowledged.
- **FR-3.3 Renal Check:** If `[Renal Impairment]` is toggled **ON** and $Drug.renalSafe == \text{false}$, the app **SHALL** display an **AMBER WARNING** banner with dose adjustment guidance.
- **FR-3.4 Max Dose Cap Alert:** When $D_{raw} \ge M_{single}$, the UI **SHALL** display a **"MAX DOSE CAPPED"** indicator alongside the capped value to prevent accidental over-prescription in larger children.

3.4 Module 4: Category Navigation & Drug Selection

- **FR-4.1:** The UI **SHALL** provide top-level category tabs for quick filtering:
    - `OPD Common` (Default)
    - `Antibiotics`
    - `ER / Emergency`
    - `Asthma / Respiratory`
    - `Favorites / Pinned`
- **FR-4.2:** Users **SHALL** be able to pin/unpin frequently used drugs to their `Favorites` tab with a single tap.
- **FR-4.3:** The app **SHALL** include a real-time instant search bar capable of filtering drugs by Generic English name (e.g., *Amoxicillin*) and Thai name (e.g., *อะม็อกซี่ซิลลิน*).

3.5 Module 5: Concentration Selection

- **FR-5.1:** Each drug card **SHALL** display its default concentration label (e.g., `120 mg/5 mL`).
- **FR-5.2:** If multiple concentrations exist for a drug, the card **SHALL** provide an inline dropdown/toggle allowing the user to switch preparations instantaneously.
- **FR-5.3:** Selecting a new concentration **SHALL** immediately recalculate the volume (V) without affecting the weight input or other drugs.

4. External Interface Requirements

4.1 User Interface (UI) Requirements

- **Color System:**
    - **Primary Action:** Medical Blue (`#1E88E5`)
    - **Safety Red Flag:** Crimson Red (`#D32F2F`)
    - **Caution / Warning:** Amber Yellow (`#FFA000`)
    - **Success / Normal:** Emerald Green (`#2E7D32`)
- **Typography:** Clean sans-serif fonts (e.g., Inter / Sukhumvit Set) with high contrast for rapid legibility under dim hospital lighting.
- **Layout:** Single-screen dashboard layout with dynamic scrolling drug cards below fixed top controls.

4.2 Software Interfaces

- **Local Storage Interface:** Interfacing with embedded NoSQL DB (Isar / Hive) for fetching drug schemas and user preference states.
- **Asset Manager:** Local file reader interface for reading bundled `drugs_v1.json` on cold boot.

4.3 Hardware Interfaces

- Supported Screen Sizes: Handheld smartphones (4.7 inches to 6.8 inches) and medical tablets (8.0 inches to 12.9 inches).
- Orientation: Portrait mode prioritized for handheld single-thumb operation.

5. Non-Functional Requirements

5.1 Performance Requirements

Metric	Target Requirement	Strict Upper Limit
Weight Input Reaction Time	$\le 10\text{ ms}$	$\le 50\text{ ms}$

Metric	Target Requirement	Strict Upper Limit
Cold App Launch Time	$\le 800\text{ ms}$	$\le 1200\text{ ms}$
Warm App Launch Time	$\le 200\text{ ms}$	$\le 400\text{ ms}$
Scroll Frame Rate	60 fps continuous	50 fps minimum
Memory Consumption	$\approx 35\text{ MB}$	$\le 60\text{ MB}$

5.2 Safety & Reliability Requirements

- **Calculation Precision:** Floating-point arithmetic must maintain accuracy to 4 decimal places internally before rendering.
- **Zero Failure Rate on Red Flags:** All safety flag conditions (Age, $G6PD$, $Renal$, $MaxCap$) must execute 100% deterministically before UI rendering occurs.
- **Data Validation:** Negative weight inputs or non-numeric characters must be blocked at input level.

5.3 Security & Compliance Requirements

- **Zero PHI Storage:** The system shall not log, store, or transmit patient names, weight history, or clinical choices.
- **Offline Independence:** No external network requests allowed during dosage calculation to ensure uninterrupted operation during network outages.

5.4 Usability & Accessibility Requirements

- **Click Count Target:** Maximum of 1 tap required (entering weight) to view dosages for standard OPD drugs.
- **Touch Targets:** Minimum touch target size of $48 \times 48\text{ dp}$ for all interactive buttons and toggles.

6. Verification & Traceability Matrix

Requirement ID	Description	Verification Method	Acceptance Criteria
FR-1.1	Single Weight Input Field	Inspection / Test	Field visible on cold launch; accepts numeric values.
FR-1.6	Real-time Update $\le 100\text{ ms}$	Automated UI Benchmark	UI recalculates and renders all cards in $< 50\text{ ms}$ upon typing.
FR-2.2	Max Single Dose Cap	Unit Test	Patient 60 kg on Paracetamol (10 mg/kg) yields capped 500 mg instead of 600 mg.
FR-3.1	Age Restriction Flag	Unit Test & UI Test	Setting age to 14 months for Chlorpheniramine displays Red Flag warning.
FR-3.2	G6PD Warning Trigger	Unit Test & UI Test	Toggling G6PD ON displays red warning on unsafe antimalarial / antibiotic cards.
NFR-5.3	Offline Execution	Integration Test	All functions remain fully operational in Airplane Mode.

7. Medical Disclaimer Specification

The application **MUST** display a non-intrusive legal disclaimer during first launch:

> **Medical Disclaimer:** *KidDose / DoseDek is designed as a clinical decision-support tool for licensed healthcare professionals. It does not replace clinical judgment. Prescribers are solely responsible for verifying final dosages against institutional protocols and patient specifics.*